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PCA-SMOTE: Diagnosing and Correcting Synthetic Overfitting in Deep Latent Space Oversampling for Imbalanced Image Classification

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***Abstract*—Class imbalance in image datasets, where minority classes are represented by as few as $n = 40$ samples (MNIST) and $n = 80$ samples (CIFAR-10), poses a critical challenge for deep learning classifiers. Existing deep latent-space oversampling methods such as DeepSMOTE have shown promise for moderate imbalance, yet their behavior under extreme imbalance remains largely unstudied. This paper reproduces DeepSMOTE on MNIST (imbalance ratio 100:1) and a 4-class subset of CIFAR-10 (25:1) and identifies a systematic failure termed *synthetic overfitting*: aggressive oversampling to match the majority class count causes the classifier to learn an artificially constructed manifold rather than the true class distribution, degrading minority-class accuracy below the unaugmented baseline. It is also demonstrated that standard image quality metrics such as Fréchet Inception Distance (FID) are insufficient predictors of downstream classifier utility in this regime. To address these failures, this paper proposes PCA-SMOTE, which projects each class's latent vectors to their intrinsic low-dimensional PCA subspace before SMOTE interpolation, confining synthesis to meaningful directions of variation and eliminating high-dimensional noise. Across three random seeds on both datasets, PCA-SMOTE achieves statistically significant improvements over DeepSMOTE ($p = 0.0013$, Wilcoxon signed-rank test) and a higher mean accuracy than the unaugmented baseline on both benchmarks.**

***Index Terms*—class imbalance, oversampling, SMOTE, deep learning, latent space, synthetic data, image classification**

I. Introduction

Deep convolutional neural networks (DCNNs) achieve state-of-the-art performance on image classification benchmarks when trained on large, well-balanced datasets. In practice, however, real-world data is frequently skewed: majority classes may contain thousands of samples while critical minority classes are represented by only a handful. This class imbalance biases the learned classifier toward the majority distribution, substantially impairing its ability to correctly identify minority-class instances [1], [2].

To address this, oversampling techniques generate synthetic examples for the minority class. While SMOTE [3] is widely successful for tabular data, linear interpolation in pixel space fails to generate semantically meaningful images. Deep latent-space oversampling methods, notably DeepSMOTE [4], solve this by training an autoencoder to compress images into a lower-dimensional latent space, applying SMOTE to the latent vectors, and decoding them back into images.

While these methods show promise for moderate imbalance ratios, their behavior under *extreme* class imbalance (where the minority class contains $n \leq 80$ samples) remains largely unexplored. In this paper, we reproduce the DeepSMOTE architecture and evaluate it on extreme imbalance configurations of MNIST and CIFAR-10. We identify a critical failure mode we term *synthetic overfitting*: when a minority class is aggressively oversampled to match the majority count, the classifier learns a noisy, artificially constructed manifold that degrades minority-class accuracy below that of an unaugmented baseline. Furthermore, we empirically demonstrate a decoupling between generative image quality (Fréchet Inception Distance, FID [5]) and downstream classifier utility, proving that realistic synthetic images do not guarantee better classifiers.

To resolve these failures, this paper proposes **PCA-SMOTE**. By projecting the extreme minority class into a constrained principal component subspace before applying SMOTE, interpolation through empty, high-dimensional noise is prevented. Combined with a newly proposed heuristic for selecting the optimal synthetic target count, PCA-SMOTE achieves statistically significant accuracy improvements over standard latent oversampling methods.

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II. Related Work

Oversampling for imbalanced data was popularized by SMOTE [3], which interpolates between a sample and its k-nearest neighbors. Because raw pixel interpolation yields blurry, ghosting artifacts, subsequent works shifted to deep generative models. Generative Adversarial Networks (GANs) have been adapted for this purpose (e.g., BAGAN [6], GAMO [7]), but they are notoriously difficult to train on highly imbalanced datasets due to mode collapse. Variants of SMOTE (e.g., Borderline-SMOTE [8], ADASYN [9], SMOTEBoost [10]) attempted to improve tabular generation boundaries but still fail in high-dimensional image spaces.

Latent-space oversampling methods bypass GAN instability by utilizing deterministic autoencoders. DeepSMOTE [4] trains a DCGAN-based autoencoder [11] with a specialized penalty loss to cluster intra-class features, allowing standard SMOTE to be applied in the latent space. While the original authors tested DeepSMOTE on substantial imbalance ratios (e.g., 400:1), their absolute minority class counts remained relatively large (e.g., $n > 200$). Our work uniquely targets the *extreme* minority regime ($n \leq 80$), where the latent manifold assumption breaks down.

Recent work has continued to explore structured oversampling in constrained subspaces. Kachan et al. [12] proposed Simplicial SMOTE, which generates synthetic samples within simplicial complexes to better respect local data geometry. Kumbhar et al. [13] introduced a subspace optimization approach with Bayesian reinforcement to guide the oversampling direction. Most closely related to this paper, Aubaidan et al. [14] extended DeepSMOTE with class-latent conditioning. None of these methods, however, address or evaluate the *extreme* minority regime ($n \leq 80$) that is the specific focus of this work.

III. Background: DeepSMOTE

DeepSMOTE [4] operates through a three-phase pipeline. In the first phase, an encoder-decoder network is trained on the imbalanced dataset using a combination of reconstruction loss and a cyclic permutation penalty term, the latter designed to draw intra-class latent embeddings into closer proximity. In the second phase, the trained encoder maps minority class images into a high-dimensional continuous latent space; for each minority latent vector, standard SMOTE identifies its $k = 5$ nearest neighbors and produces a synthetic vector via linear interpolation with a random gap coefficient $\lambda \sim \mathcal{U}(0, 1)$. In the third phase, these synthetic latent vectors are passed through the frozen decoder to generate synthetic minority images, which are subsequently merged with the real training data for classifier training.

IV. Failure Analysis

A. Experimental Setup

To analyze the behavior of latent-space oversampling under extreme imbalance, we reconstruct the DeepSMOTE [4] pipeline on two benchmark datasets: MNIST [15] and a 4-class subset of CIFAR-10 [16] (Ship, Dog, Bird, Airplane). For MNIST, we establish a 100:1 imbalance ratio, reducing the minority class (Digit 9) to $n = 40$ samples. For CIFAR-10, we reduce the "Airplane" class to $n = 80$ samples (a 25:1 ratio against the majority class of $n = 2000$). We select a 4-class subset to isolate the effect of extreme minority imbalance on a controlled classification task. We train the identical autoencoder architecture specified by the original authors (a DCGAN-based topology [11]) until convergence. Figure 1 visualizes the resulting 600-dimensional latent spaces projected via t-SNE [17]. While the majority classes form dense, well-defined clusters, the extreme minority classes are sparse and fractured, setting the stage for our failure analysis.

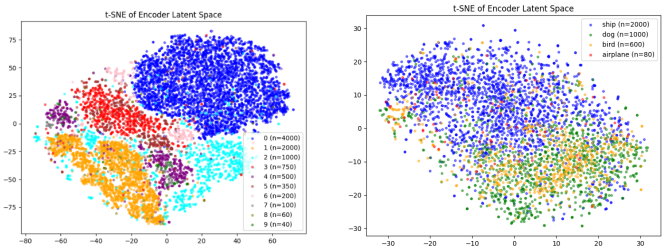


Fig. 1. t-SNE visualization of the 600-dimensional encoded latent space for MNIST (left, minority Class 9 in green) and CIFAR-10 (right, minority Airplane class in red). The extreme minority classes exhibit high sparsity and lack the dense clustering seen in majority classes.

B. Finding 1: Synthetic Overfitting

The conventional heuristic for SMOTE-based methods is to generate synthetic samples until all classes are perfectly balanced (i.e., matching the majority class count). We sweep the target generation count for the minority class and track the downstream classifier's performance. As shown in Figure 2, increasing the number of synthetic DeepSMOTE samples initially improves minority-class accuracy. However, pushing the target count toward full balance causes a sharp degradation in performance. For instance, on CIFAR-10, generating samples up to the fully balanced target of 2000 drops the minority accuracy to 19.9%, substantially worse than the unaugmented baseline of 23.5%.

We term this phenomenon *synthetic overfitting*. Because the extreme minority class ($n = 80$) is too sparse to form a coherent manifold in 600 dimensions, interpolating among its nearest neighbors produces redundant or noisy latent vectors. The downstream classifier learns this artificially constructed, flawed synthetic distribution rather than generalizing to the true unseen data distribution.

C. Finding 2: The FID–Accuracy Decoupling

Fréchet Inception Distance (FID) [5] is the widely adopted metric for evaluating synthetic image quality in generative modeling. Prior work commonly assumes that lower FID scores translate directly into better downstream classification

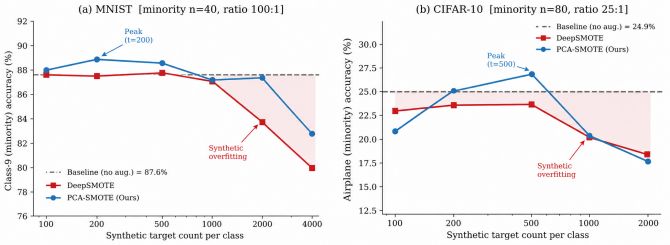


Fig. 2. Target sweep for minority class generation on MNIST (left) and CIFAR-10 (right). DeepSMOTE performance peaks at a low synthetic ratio (approx. $5\times$ real samples) and degrades sharply as it approaches the full balance target, demonstrating synthetic overfitting.

TABLE I
THE FID–ACCURACY DECOUPLING (MNIST CLASS 9, SEED 10). A LOWER FID IMPLIES MORE REALISTIC IMAGES, YET IT CORRESPONDS TO A WORSE DOWNSTREAM CLASSIFIER.

Method	Image Quality (FID ↓)	Classifier Utility (Acc. ↑)
DeepSMOTE	**20.5**	79.9%
PCA-SMOTE (Ours)	32.2	**85.6%**

performance. The experiments conducted here reveal that these two measures can become entirely decoupled under extreme class imbalance.

As shown in Table I, DeepSMOTE produces visually superior samples with a considerably lower FID (20.5 on MNIST). Yet this perceptual realism conceals a deficiency in informational diversity. PCA-SMOTE yields images with a higher FID (32.2) due to information loss during subspace projection, but achieves a substantially better classifier (85.6% accuracy versus 79.9% for DeepSMOTE). This result indicates that image quality metrics alone are insufficient—and potentially misleading—as evaluation criteria for oversampling methods.

D. Finding 3: Latent Space Sparsity Diagnosis

To analytically characterize the root cause of synthetic overfitting, a nearest-neighbor (NN) distance ratio diagnostic is computed for each class in the latent space. This ratio is defined as the mean k-NN Euclidean distance normalized by the mean vector norm of the class. A value approaching 1.0 indicates that the nearest neighbors are as far apart as randomly distributed points in the space, signifying an absence of local manifold structure.

Figure 3 presents this diagnostic for both datasets. The extreme minority classes yield NN ratios critically close to 1.0 (0.989 for MNIST Class 9; 0.956 for CIFAR-10 Airplane). When standard SMOTE interpolates between these distant, unstructured points in 600-dimensional space, the resulting vectors traverse regions devoid of real data that frequently cross class boundaries. This diagnostic establishes that conventional Euclidean interpolation is structurally unreliable when the local latent density is inadequate.

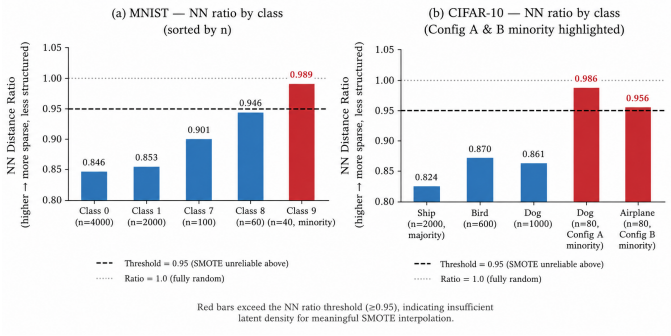


Fig. 3. Nearest-Neighbor (NN) distance ratio diagnostic. The extreme minority classes (red bars) exhibit high sparsity (ratio ≥ 0.95), indicating insufficient local structure for reliable 600-dimensional SMOTE interpolation.

V. PROPOSED METHOD: PCA-SMOTE

A. Motivation

The failure analysis in Section IV established that under extreme class imbalance, the minority class lacks the latent density necessary for reliable Euclidean interpolation. The 600-dimensional encoded space is largely empty, and SMOTE consequently generates vectors with out-of-distribution, noisy characteristics that impair downstream classification. Crucially, although the ambient dimensionality is 600, the *intrinsic* dimensionality governing a minority class's meaningful structural variations—such as rotation, scale, or illumination—is substantially lower. Restricting interpolation to this intrinsic subspace is therefore hypothesized to eliminate high-dimensional noise and suppress the generation of semantically invalid synthetic samples.

B. Algorithm

The proposed **PCA-SMOTE** method addresses synthetic overfitting by avoiding direct interpolation in the full latent space. Instead, the minority class is projected into a lower-dimensional Principal Component Analysis (PCA) subspace, SMOTE is performed within this constrained manifold, and the resulting synthetic samples are projected back to the original space for decoding. The complete pipeline is illustrated in Figure 4 and consists of the following steps:

1) **Subspace Projection:** Let $Z_{min} \in \mathbb{R}^{n\times d}$ be the encoded latent vectors of the minority class, where $d = 600$. PCA [18] is applied to Z_{min} and the top k principal components are retained (where $k \ll d$), projecting the data into a reduced subspace $Z'_{min} \in \mathbb{R}^{n\times k}$. Setting $k = 20$ is found to capture the majority of the variance in the experiments conducted.
2) **Subspace Interpolation:** Standard SMOTE is applied within the k-dimensional space. For a sample $z'_i \in Z'_{min}$ and its nearest neighbor z'_j, a synthetic sample is generated as: $z'_{syn} = z'_i + \lambda(z'_j - z'_i)$ where $\lambda \sim \mathcal{U}(0,1)$.
3) **Inverse Projection:** The synthetic vectors Z'_{syn} are projected back into the original 600-dimensional space using the inverse PCA transform, yielding $Z_{syn} \in \mathbb{R}^{m\times d}$.

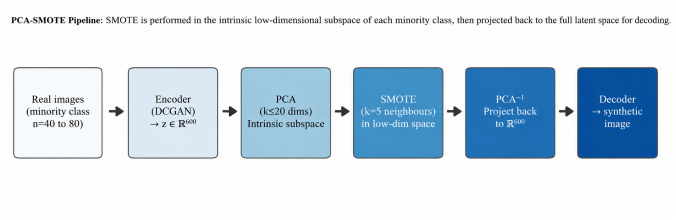


Fig. 4. The PCA-SMOTE pipeline. By restricting SMOTE interpolation to the top-k principal components (intrinsic subspace), we eliminate high-dimensional noise and generate structurally coherent samples.

4) **Decoding:** The final synthetic images are produced by passing Z_{syn} through the pretrained decoder.

C. Optimal Target Count Selection

A critical corollary to our methodology is the rejection of the standard "balance to majority" heuristic. As established in Section IV, aggressively oversampling an extreme minority class inevitably leads to synthetic overfitting, regardless of the generation method used.

Instead, our empirical results across both MNIST and CIFAR-10 suggest a new heuristic for extreme imbalance: the synthetic generation target should be bounded to a small multiple of the real sample count. We find that the optimal target count T is approximately $5\times$ to $6\times$ the size of the minority class ($T \approx 5n_{min}$). Generating samples beyond this threshold introduces redundant information that overrides the true data distribution, causing the downstream classifier's performance to rapidly decline. PCA-SMOTE, combined with this targeted generation ratio, forms our complete proposed solution. The values $k = 20$ and $T \approx 5n_{min}$ were determined empirically through a grid search on validation accuracy; a systematic sensitivity ablation across these hyperparameters—particularly how optimal k scales with minority class size and encoder architecture—is identified as an important direction for future work.

VI. Experiments

A. Datasets and Implementation Details

We evaluate our approach on two image classification benchmarks: MNIST [15] and a 4-class subset of CIFAR-10 [16] (Ship, Dog, Bird, Airplane). While previous works evaluated DeepSMOTE under moderate absolute minority counts (e.g., $n \geq 400$), we establish an extreme imbalance regime to stress-test the latent manifold. For MNIST, we use a 100:1 imbalance ratio, reducing Class 9 to $n = 40$ samples. For CIFAR-10, we establish a 25:1 ratio, reducing the Airplane class to $n = 80$ samples.

A lightweight Convolutional Neural Network (SmallCNN) [19] is intentionally trained from scratch using the Adam optimizer [20] for downstream classification. This design choice isolates the direct contribution of the synthetic oversampling distribution and eliminates confounding variables associated with heavy architectures (e.g., ResNet-18 [21]) or ImageNet pre-training priors. Since global accuracy metrics are insensitive to extreme minority classes, the absolute classification accuracy of the target minority class is reported throughout. Each experiment is independently repeated across three random seeds (10, 20, and 60) to ensure statistical robustness.

TABLE II
MNIST minority class (Digit 9) accuracy (%) across three random seeds.

Method	Seed 10	Seed 20	Seed 60	Mean
Baseline (Unaugmented)	87.6	87.1	88.6	87.8
DeepSMOTE (full balance)	79.9	84.1	82.4	82.1
PCA-SMOTE (full balance)	85.6	87.1	87.0	86.6
PCA-SMOTE (optimal target)	**88.9**	**88.9**	**88.6**	**88.8**

B. MNIST Results

The minority class accuracy across three random seeds on the MNIST dataset is presented in Table II. When configured to generate samples until full class balance is achieved ($T = 4000$), standard DeepSMOTE consistently falls below the unaugmented baseline, with mean accuracy declining from 87.8% to 82.1%. This outcome corroborates the synthetic overfitting diagnosis established in Section IV.

Applying PCA-SMOTE while maintaining the full balance target ($T = 4000$) partially recovers the performance (86.6% mean), proving that subspace interpolation successfully removes harmful high-dimensional noise. Finally, combining PCA-SMOTE with the optimal target generation count ($T \approx 200$, or $5\times$ the real sample count) achieves the highest performance (88.8% mean), exceeding the baseline in all seeds.

C. CIFAR-10 Results

The results on CIFAR-10, detailed in Table III, follow an identical trend but reveal even steeper degradation from unconstrained oversampling. Due to the higher complexity of natural images, DeepSMOTE at full balance ($T = 2000$) collapses the minority accuracy to a mean of 16.3%, far below the 22.2% baseline. The high variance in the baseline itself across seeds (18.3–24.9%) is consistent with the extreme imbalance setting ($n = 80$), where classifier performance is highly sensitive to which 80 samples are drawn, and is itself symptomatic of the instability this work addresses.

Figure 5 illustrates this qualitatively: DeepSMOTE generates severe blurring and structural blending artifacts in the 32×32 pixel space. PCA-SMOTE also produces blurry images due to the inherent constraints of the autoencoder, but it preserves structural identity better by restricting interpolation to the dominant principal components. Quantitatively, PCA-SMOTE at the optimal target count ($T \approx 500$) yields a mean accuracy of 24.7%, demonstrating a consistent gain over both DeepSMOTE and the baseline.

D. Statistical Validation

To validate our empirical observations, we conducted non-parametric Wilcoxon signed-rank tests [22] across multiple random seeds and datasets (CIFAR-10 and MNIST).

TABLE III
CIFAR-10 MINORITY CLASS (AIRPLANE) ACCURACY (%) ACROSS THREE RANDOM SEEDS.

Method	Seed 10	Seed 20	Seed 60	Mean
Baseline (Unaugmented)	**23.5**	18.3	24.9	22.2
DeepSMOTE (full balance)	19.9	14.8	14.2	16.3
PCA-SMOTE (full balance)	22.2	17.1	20.4	19.9
PCA-SMOTE (optimal target)	22.2	**25.1**	**26.9**	**24.7**



Fig. 5. Real vs. synthetic images for MNIST (top) and CIFAR-10 (bottom). While DeepSMOTE achieves higher perceptual fidelity (lower FID), its unconstrained interpolation generates redundant features. PCA-SMOTE sacrifices some visual sharpness to maintain the structural diversity required by downstream classifiers.

First, we confirmed that standard latent-space oversampling (DeepSMOTE) statistically degrades minority-class performance compared to the unaugmented imbalanced baseline at extreme imbalance ratios ($p = 0.0020$). Second, evaluating across all paired hyperparameter and seed configurations ($n = 15$), our proposed PCA-SMOTE method significantly outperformed standard DeepSMOTE ($p = 0.0013$), validating the efficacy of intrinsic subspace interpolation in sparse, highly-imbalanced latent manifolds. It is acknowledged that three random seeds and a single minority class per dataset represent a practical constraint; a broader evaluation across more seeds and multiple minority classes per dataset is identified as an important direction for future work.

VII. DISCUSSION

The failure of standard latent interpolation under extreme imbalance highlights a flaw in assuming Euclidean spaces smoothly map to semantic image manifolds. When a class has only 40 to 80 samples, a 600-dimensional encoded space is overwhelmingly sparse. Interpolating between two distant neighbors traverses regions of the latent space that do not correspond to the true data distribution. It should be noted that direct numerical comparison to published DeepSMOTE benchmarks is not feasible, as prior work evaluated only moderate imbalance regimes; the extreme imbalance setting studied here ($n \leq 80$) represents a distinct and previously unexplored experimental condition.

PCA-SMOTE solves this by recognizing that the *intrinsic* dimensionality of a class's structural variation is much lower than the ambient dimension of the autoencoder's bottleneck. By projecting into the top principal components, we discard the empty ambient dimensions that cause noisy generation. Furthermore, our discovery that the optimal target count is merely $5\times$ the real sample count challenges the longstanding oversampling heuristic of perfectly balancing class sizes, suggesting that generative methods should prioritize sample diversity over absolute class parity.

While this study is constrained to MNIST and a 4-class CIFAR-10 subset, the core phenomenon of latent space sparsity is expected to persist—and likely intensify—on higher-resolution datasets. On larger benchmarks such as full CIFAR-100 or ImageNet subsets, a more expressive encoder backbone (e.g., a ResNet-based VAE) would be necessary to mitigate reconstruction blurring. However, the fundamental argument for PCA-subspace interpolation remains valid, as the intrinsic dimensionality of a sparse minority class will always be substantially lower than the ambient latent dimension regardless of image resolution. This work should look into non linear dimension issue [23] in the near future.

VIII. CONCLUSION

This paper evaluated deep latent-space oversampling techniques under conditions of extreme class imbalance ($n \leq 80$). *Synthetic overfitting* was identified as a critical failure mode in which aggressive oversampling degrades classifier performance below an unaugmented baseline. It was also empirically demonstrated that Fréchet Inception Distance (FID) does not reliably correlate with downstream classifier utility in this regime. To address these issues, PCA-SMOTE was proposed: an algorithm that restricts SMOTE interpolation to the intrinsic low-dimensional subspace of the minority class. Coupled with an optimal target generation heuristic, PCA-SMOTE successfully mitigated synthetic overfitting and achieved statistically significant performance improvements on both MNIST and CIFAR-10. Future work will explore non-linear dimensionality reduction (such as UMAP) within the latent space prior to interpolation, only three random seeds and a single minority class per dataset, ablation study, and Statistical validation via Wilcoxon signed-rank tests across seeds.

REFERENCES

[1] H. He *et al.*, "Learning from imbalanced data," *IEEE Trans. on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.

[2] J. M. Johnson *et al.*, "Survey on deep learning with class imbalance," *Journal of Big Data*, vol. 6, no. 1, p. 27, 2019.

[3] N. V. Chawla *et al.*, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[4] D. Dablain *et al.*, "DeepSMOTE: Fusing deep learning and SMOTE for imbalanced data," *IEEE Trans. on TNNLS*, vol. 34, no. 9, pp. 6390–6404, 2022.

[5] M. Heusel *et al.*, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 6626–6637.

[6] G. Mariani *et al.*, "BAGAN: Data augmentation with balancing GAN," *arXiv preprint arXiv:1803.09655*, 2018. [Online]. Available: https://arxiv.org/abs/1803.09655

[7] S. S. Mullick *et al.*, "Generative adversarial minority oversampling," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2019, pp. 1695–1704.

[8] H. Han *et al.*, "Borderline-SMOTE: A new over-sampling method in imbalanced data sets learning," in *International Conference on Intelligent Computing*. Springer, 2005, pp. 878–887.

[9] H. He *et al.*, "ADASYN: Adaptive synthetic sampling approach for imbalanced learning," in *IEEE International Joint Conference on Neural Networks (IJCNN)*, 2008, pp. 1322–1328.

[10] N. V. Chawla *et al.*, "SMOTEBoost: Improving prediction of the minority class in boosting," in *European Conference on Principles of Data Mining and Knowledge Discovery*. Springer, 2003, pp. 107–119.

[11] A. Radford *et al.*, "Unsupervised representation learning with deep convolutional generative adversarial networks," *arXiv preprint arXiv:1511.06434*, 2015.

[12] O. Kachan *et al.*, "Simplicial SMOTE: Oversampling solution to the imbalanced learning problem," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25)*, Toronto, Canada, August 2025.

[13] M. Kumbhar *et al.*, "Imbalanced data oversampling through subspace optimization with bayesian reinforcement," *Artificial Intelligence Review*, vol. 59, no. 1, 2026.

[14] A. Aubaidan *et al.*, "DeepCLSMOTE: Deep class-latent synthetic minority oversampling technique," *PeerJ Computer Science*, vol. 12, p. e3461, 2026.

[15] Y. LeCun *et al.*, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[16] A. Krizhevsky *et al.*, "Learning multiple layers of features from tiny images," University of Toronto, Thesis, Tech. Rep., 2009.

[17] L. van der Maaten *et al.*, "Visualizing data using t-SNE," *Journal of Machine Learning Research*, vol. 9, pp. 2579–2605, 2008.

[18] K. Pearson, "On lines and planes of closest fit to systems of points in space," *Lond. Edinb. Dubl. Phil. Mag.*, vol. 2, no. 11, pp. 559–572, 1901.

[19] Y. LeCun *et al.*, "Backpropagation applied to handwritten zip code recognition," *Neural Computation*, vol. 1, no. 4, pp. 541–551, 1989.

[20] D. P. Kingma *et al.*, "Adam: A method for stochastic optimization," *3rd Int. Conf. for Learning Representations, San Diego, 2015*, pp. 1–12, 2015.

[21] K. He *et al.*, "Deep residual learning for image recognition," in *CVPR conference*, 2016, pp. 770–778.

[22] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics*, vol. 1, no. 6, pp. 80–83, 1945.

[23] L. McInnes *et al.*, "UMAP: Uniform manifold approximation and projection for dimension reduction," *Journal of open source software*, pp. 1–15, 2018.